University of the Witwatersrand
School of Computer Science and Applied Mathematics
COMS4070A & COMS7081A:
Computational Neuroscience
Population Coding, Dynamics and Memory Tutorial
(Lectures 4–6)

How to use this tutorial

This tutorial combines **single best answer and written questions** (concepts and derivations) with a **companion notebook** (`tutorial_notebook_solutions.ipynb`). These notebook tasks are entirely optional extension material and will not be in the test! The aim is to help build intuition for derivations you do on paper, so I have filled in the programming answers already. Where a question is marked [**Notebook Task X**], do the written part first and then use the notebook to confirm your answer.

The written questions give an example of the types of questions you can expect in a test or exam. Some may be repetitive, to help calibrate to the different ways the same concept may be asked. Marks are an indication only.

1 Warm up: Quiz style single best answer questions

1. [**Multiple Choice**] For a Poisson spike count with rate λ, the variance of the count is:

   a) λ b) λ^2 c) $\sqrt{\lambda}$ d) 1

2. [**True/False**] The maximum likelihood estimate for a linear neuron with Gaussian noise of constant variance is the least-squares solution.

   a) True b) False

3. [**True/False**] The spike-triggered average recovers a neuron's linear filter for any stimulus distribution.

   a) True b) False

4. [**Multiple Choice**] In the GLM $r_t \sim \text{Poisson}(f(\mathbf{s}_t \cdot \mathbf{w}))$, the main role of the non-linearity f is to:

   a) reduce the dimensionality of the stimulus.
   b) map the linear drive to a non-negative firing rate.
   c) add noise to the response.
   d) make the log-likelihood linear in $\mathbf{w}$.

5. [**Multiple Choice**] For $\tau\dot{r} = -r + wr + I$ with $w = 0.75$, the effective time constant is:

   a) $\tau/4$ b) 0.75τ c) 4τ d) undefined (unstable)

6. [**Multiple Choice**] A fixed point has Jacobian eigenvalues $\lambda = -0.5 \pm 2i$. Nearby trajectories:

   a) grow without oscillating.
   b) decay without oscillating.
   c) spiral inwards (damped oscillations).
   d) form a stable limit cycle.

7. **[True/False]** In a balanced network, excitatory and inhibitory inputs are each individually small.

   a) True b) False

8. **[True/False]** In an inhibition-stabilised network, increasing external input to the inhibitory population can decrease inhibitory firing at steady state.

   a) True b) False

9. **[Multiple Choice]** For random dense patterns in a classical Hopfield network with N neurons, the maximum number of patterns that can be reliably stored is approximately:

   a) 0.138 b) $0.138N$ c) $0.138N^2$ d) 2^N

10. **[Matching Pairs]** Match each variable to the attractor geometry best suited to represent it: (i) horizontal eye position, (ii) head direction, (iii) one of several stored odours.

    a) point attractors b) line attractor c) ring attractor

2 Neural coding and population representations (Lecture 4)

2.1 Tuning curves and the spike-triggered average

A.1 Define a *tuning curve*. Name four properties of a neuron's code that can be read off its tuning curve. [4 Marks]

A.2 Explain how the spike-triggered average (STA) is computed. Why do stimulus features that are unrelated to spiking "average away"? What assumption about the stimulus makes the STA recover the neuron's linear filter? [4 Marks]

A.3 Distinguish *encoding* from *decoding*. In which sense is the STA used for each? [3 Marks]

A.4 Why is the Poisson distribution a natural first choice for spike counts in a time window? State one way in which real neurons deviate from it. [3 Marks]

2.2 From a neuron with a constant rate to MLE

A neuron fires at an unknown constant rate λ. We count its spikes $n_1, \ldots, n_n$ in n independent 1 s windows, $n_i \sim \text{Poisson}(\lambda)$, and write $S = \sum_i n_i$.

A.5 **[Notebook Task A1 (extension)]** Spike counts are often modelled as Gaussian.

(a) Derive the MLEs $\hat{\mu}$ and $\hat{\sigma}^2$ for i.i.d. Gaussian data, following the lecture's approach for the Poisson MLE if you get stuck. [4 Marks]

(b) If the counts are actually Poisson, what values for mean and variance do you expect the counts to approach? [1 Mark]

(c) Given that the two models fit high-rate data about equally well, why might we still prefer the Poisson model? (Bonus points if you know *why* they both fit high rates well, but this is not expected!) [2 Marks]

Now suppose a neuron responds to contrast $s_i \geq 0$ on trial i ($i = 1, \ldots, n$) with a linear rate $\lambda_i = \theta s_i$.

A.6 **[Notebook Task A2–A3]** If the spike count is $n_i \sim \text{Poisson}(\theta s_i)$, write the likelihood $P(\mathbf{n} \mid \mathbf{s}, \theta)$ and state the assumption needed to write it as a product over trials. Derive the maximum likelihood estimate $\hat{\theta}_{\text{ML}}$ and verify it is a maximum. [6 Marks]

2.3 Generalised linear models

A.7 Consider the linear-Gaussian GLM $\mathbf{r} = S\mathbf{w} + \boldsymbol{\epsilon}$, $\boldsymbol{\epsilon} \sim \mathcal{N}(0, \sigma^2 I)$. Derive $\hat{\mathbf{w}} = (S^\top S)^{-1} S^\top \mathbf{r}$. When does this quantity fail to exist? [4 Marks]

A.8 **[Notebook Task A4]** For the Poisson GLM $r_t \sim \text{Poisson}(\Delta\, \lambda_t)$ with $\lambda_t = \exp(\mathbf{s}_t \cdot \mathbf{w})$: [7 Marks]

(a) Write down the log-likelihood (dropping terms that do not depend on $\mathbf{w}$).

(b) Compute its gradient with respect to $\mathbf{w}$.

2.4 Populations and latent variables

A.9 What is a population code? Give two reasons why a population of noisy neurons with different tuning curves can represent a stimulus better than any single neuron. [4 Marks]

A.10 In PCA we write $X \approx ZW^\top$ with $X \in \mathbb{R}^{T \times N}$ and $Z \in \mathbb{R}^{T \times K}$, $K \ll N$. Explain what Z and W represent in terms of neural activity. Give an example of a latent variable that the experimenter does not control but which might dominate the population activity. [4 Marks]

3 Recurrent dynamics and E–I balance (Lecture 5)

3.1 From spiking networks to rate models

B.1 A network of N conductance-based neurons has $2N$ state variables and products like $g_j(t)V_i(t)$. Explain what we gain and what we lose by replacing it with the rate model $\tau \frac{d\boldsymbol{r}}{dt} = -\boldsymbol{r} + \phi(W\boldsymbol{r} + \boldsymbol{I})$. [4 Marks]

B.2 Explain the role of each term in $\tau \frac{d\boldsymbol{r}}{dt} = -\boldsymbol{r} + \phi(W\boldsymbol{r} + \boldsymbol{I})$: τ, $-\boldsymbol{r}$, $W\boldsymbol{r}$, $\boldsymbol{I}$ and ϕ. [5 Marks]

3.2 A single recurrent population

Take $\phi(x) = x$ and a single population: $\tau \frac{dr}{dt} = -r + wr + I$.

B.3 Derive the fixed point r^* and the effective time constant τ_{eff} of small perturbations about it. Complete the table below and describe in words what recurrent excitation does to the network's gain and speed. [6 Marks]

w	gain r^*/I	τ_{eff}/τ
0		
0.5		
0.9		
0.99		

B.4 What happens when $w > 1$? When $w < 0$? What stops the activity of real networks from growing without bound? [4 Marks]

B.5 What happens at exactly $w = 1$? Why might this be useful, and why is it biologically delicate? [3 Marks]

3.3 The two-population E–I network

$$\tau_E \dot{r}_E = -r_E + \phi(w_{EE} r_E - w_{EI} r_I + I_E), \qquad \tau_I \dot{r}_I = -r_I + \phi(w_{IE} r_E - w_{II} r_I + I_I)$$

with all $w \geq 0$. For B.8–B.12, take $\phi(x) = x$, $\tau_E = \tau_I = 1$, and

$$w_{EE} = 2, \quad w_{EI} = 2, \quad w_{IE} = 2, \quad w_{II} = 0.5, \quad I_E = 1, \quad I_I = 0.5.$$

B.6 Draw the E–I circuit, labelling each connection with its sign. Describe the feedback loop that makes inhibition stabilising. [3 Marks]

B.7 Define a *nullcline*. Why must every fixed point lie on the intersection of the two nullclines? [2 Marks]

B.8 **[Notebook Task B1]** Derive both nullclines as functions $r_I(r_E)$ and find the fixed point. [5 Marks]

B.9 Write down the Jacobian J of this system. [4 Marks]

B.10 **[Notebook Task B2]** The eigenvalues λ of J are given by analysing the determinant of the Jacobian, i.e. since $\mathrm{tr} J = -0.5$, $\det J = -1.5 + 4 = 2.5$ and $\lambda^2 + 0.5\lambda + 2.5 = 0 \Rightarrow \lambda = -0.25 \pm i\sqrt{2.4375} \approx -0.25 \pm 1.56i$. Classify the fixed point (refer to Figure 2). From this, you can also predict the oscillation period and the decay time constant of perturbations, and check the prediction against Figure 1 (not expected for the test but included for interest). [3 Marks]

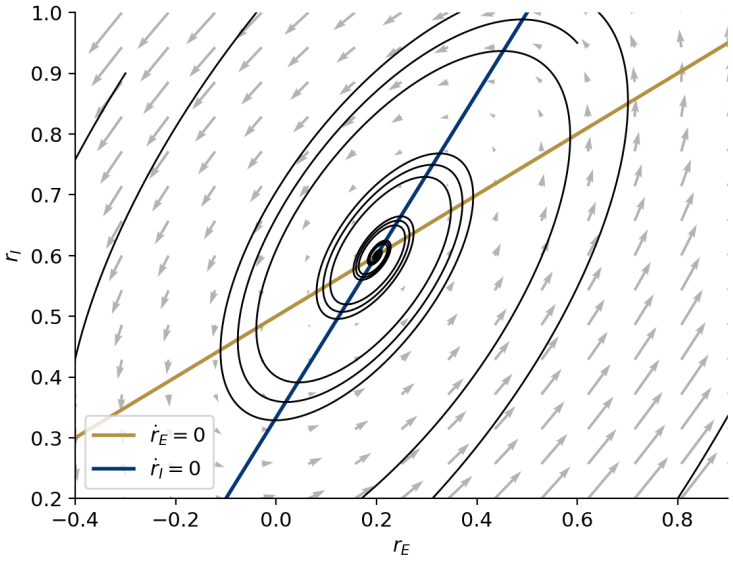


Figure 1: Phase plane of the linear E–I network in B.6–B.10: nullclines, vector field (grey) and three trajectories (black).

B.11 **Inhibition-stabilised networks.** [8 Marks] **[Notebook Task B3]**

(a) Explain in words why the excitatory population on its own (with r_I held fixed) would be unstable.

(b) Solve for the fixed point r_I^* as a function of I_I (keeping $I_E = 1$). What happens to inhibitory activity when we give the inhibitory population *more* input? Explain this "paradoxical effect" intuitively.

(c) Repeat (b) with $w_{EE} = 0.5$. What changes?

3.4 Non-linear E–I dynamics and balanced networks

B.12 Restore the non-linearity ϕ. Derive all four entries of the Jacobian of the Wilson–Cowan system at a fixed point $(r_{\mathrm{E}}^*, r_{\mathrm{I}}^*)$. [4 Marks]

B.13 Give one example of how a disturbance of E–I balance has been linked to a clinical condition, and explain it in dynamical terms. [3 Marks]

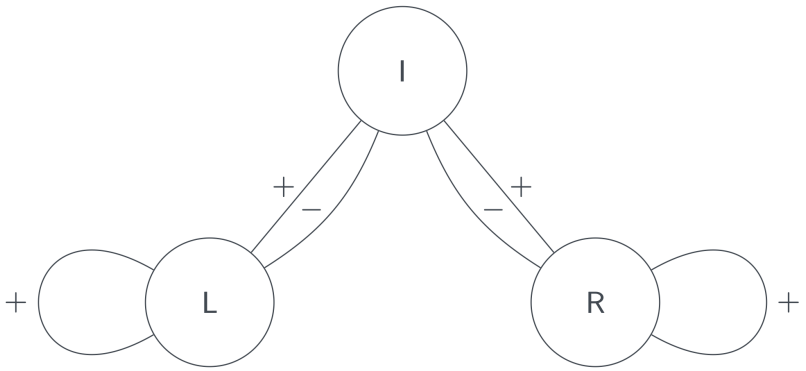
4 Attractors, memory and canonical computation (Lecture 6)

4.1 Multiple stable states

C.1 Using a sketch of $y = r$ and $y = \phi(wr + I)$ for a single population, explain why a linear ϕ gives at most one fixed point but a sigmoidal ϕ can give three. [3 Marks]

C.2 Define an *attractor* and a *basin of attraction*. Distinguish point attractors from continuous attractors and give a neural example of each of a point, a line and a ring attractor. [6 Marks]

C.3 State the three ingredients a recurrent circuit needs to implement winner-take-all decisions, and explain how the circuit shown below (from the Lecture 6 WTA slide) provides each. [3 Marks]



$$E_L \uparrow \rightarrow I \uparrow \rightarrow E_R \downarrow$$

One population wins; the other is suppressed.

4.2 Hopfield networks

Assume that neurons are binary, $s_i \in \{-1, +1\}$. We store P patterns ξ^μ with the Hebbian rule

$$J_{ij} = \frac{1}{N} \sum_{\mu=1}^{P} \xi_i^\mu \xi_j^\mu ,$$

$J_{ii} = 0$ and update $s_i \leftarrow \mathrm{sign}(h_i)$ with local field $h_i = \sum_j J_{ij} s_j$.

C.4 Interpret the Hebbian rule: for which pairs of neurons does a single pattern make J_{ij} positive, and for which negative? Why is J symmetric? [3 Marks]

C.5 Why do we set $J_{ii} = 0$? [1 Mark]

C.6 **[Notebook Task C1] By hand.** Store the two patterns ($N = 6$)

$$\xi^1 = (+1, +1, +1, -1, -1, -1), \qquad \xi^2 = (+1, +1, -1, +1, -1, -1).$$

(a) Compute the 6×6 weight matrix J (you may give $6J$).

(b) The network starts in the cue $\boldsymbol{s} = (+1, +1, +1, -1, +1, -1)$. Compute the overlap $m^\mu = \frac{1}{N} \sum_i \xi_i^\mu s_i$ for each of the two patterns.

(c) Compute all six local fields and the state after one synchronous update. Which pattern has been retrieved?

(d) Compute the energy $E(\boldsymbol{s}) = -\frac{1}{2} \sum_{i,j} J_{ij} s_i s_j$ of the cue and of the retrieved state.

[10 Marks]

C.7 Now start the network exactly in ξ^1. Compute the local fields. Which neurons are held in place least strongly, and why? Relate your answer to the overlap $\xi^1 \cdot \xi^2$. [4 Marks]

C.8 Suppose the network is exactly in memory ξ^μ. Substitute the Hebbian weights into h_i and separate the result into a *signal* term and a *crosstalk* term. Under what condition on the crosstalk is ξ^μ a fixed point? [5 Marks]

C.9 Show that if the stored patterns are mutually orthogonal ($\xi^\mu \cdot \xi^\nu = 0$ for $\mu \neq \nu$) every pattern is a fixed point for any $P < N$. Relate this to the idea of *whitening* the representation, and explain why storing many similar memories (e.g. similar faces) is hard. [5 Marks]

C.10 List three biologically unrealistic features of the classical Hopfield network and, for each, a modification that addresses it. [6 Marks]

5 List of Symbols and Equations

5.1 Neural coding (Lecture 4)

Symbol	Definition
s, $\mathbf{s}_t$	Stimulus (scalar / stimulus history vector at time t)
S	Design matrix of (shifted) stimuli
r, n	Response: rate or spike count
λ	Firing rate (Poisson mean)
θ, $\mathbf{w}$	Model parameters / linear filter
$f(\cdot)$	GLM non-linearity
Δ	Time bin width
$\boldsymbol{r} = (r_1, \ldots, r_N)$	Population response
$\boldsymbol{z}$, W	Latent variables and their mapping to neurons

Spike-triggered average: $\mathrm{STA}(\tau) = \dfrac{1}{N}\sum_{k=1}^{N} s(t_k - \tau)$

Maximum likelihood: $\hat{\theta}_{\mathrm{ML}} = \arg\max_{\theta} \prod_i P(r_i \mid s_i, \theta)$

Poisson distribution: $P(n \mid \lambda) = \dfrac{\lambda^n e^{-\lambda}}{n!}$

Linear-Gaussian GLM: $\hat{\mathbf{w}} = (S^\top S)^{-1} S^\top \mathbf{r}$

Poisson GLM log-likelihood: $\mathcal{L} = \mathbf{r}^\top \log f(S\mathbf{w}) - \Delta \mathbf{1}^\top f(S\mathbf{w})$

Bernoulli GLM log-likelihood: $\mathcal{L} = \mathbf{r}^\top \log f(S\mathbf{w}) + (\mathbf{1} - \mathbf{r})^\top \log[1 - f(S\mathbf{w})]$

PCA / linear latent model: $\boldsymbol{r}_t = W\boldsymbol{z}_t + \boldsymbol{\epsilon}_t$

5.2 Recurrent dynamics (Lecture 5)

Symbol	Definition
τ, τ_{E}, τ_{I}	Population time constants
W, w_{ab}	Recurrent weights (from population b to a)
ϕ, ϕ'	Input–output non-linearity and its slope (gain)
$\boldsymbol{r}^*$	Fixed point
J	Jacobian at a fixed point
$\lambda = \alpha \pm i\omega$	Eigenvalues of J (decay rate, angular frequency)
K	Number of inputs per neuron

Rate model: $\tau\dot{\boldsymbol{r}} = -\boldsymbol{r} + \phi(W\boldsymbol{r} + \boldsymbol{I})$

Single linear population: $r^* = \dfrac{I}{1-w}, \quad \tau_{\text{eff}} = \dfrac{\tau}{1-w}$

Wilson–Cowan Jacobian:

$$J = \begin{pmatrix} \dfrac{-1 + w_{\text{EE}}\phi'_{\text{E}}}{\tau_{\text{E}}} & \dfrac{-w_{\text{EI}}\phi'_{\text{E}}}{\tau_{\text{E}}} \\ \dfrac{w_{\text{IE}}\phi'_{\text{I}}}{\tau_{\text{I}}} & \dfrac{-1 - w_{\text{II}}\phi'_{\text{I}}}{\tau_{\text{I}}} \end{pmatrix}$$

2×2 **stability:** stable iff $\text{Tr}(J) < 0$ and $\det J > 0$; oscillatory iff $(\text{Tr}(J))^2 < 4 \det J$.

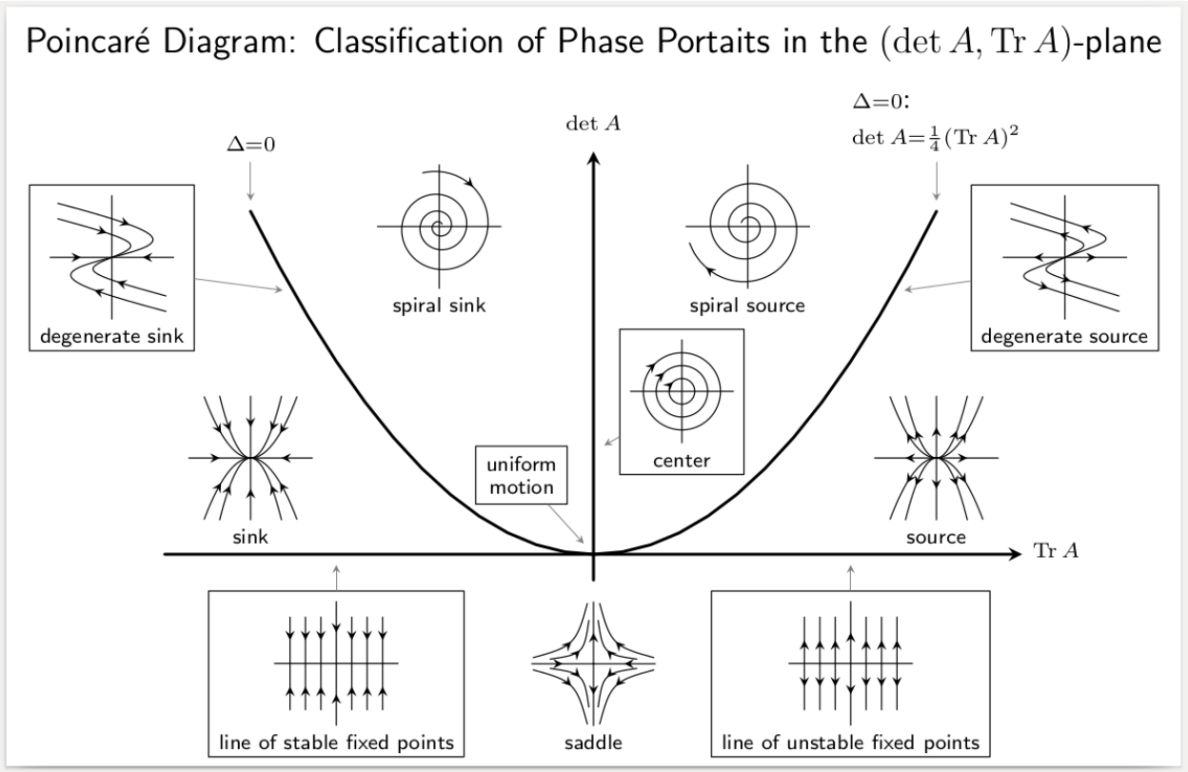


Figure 2: Poincaré diagram to aid classification of fixed points in 2D phase planes.

5.3 Attractors and Hopfield networks (Lecture 6)

Symbol	Definition
$s_i \in \{-1, +1\}$	State of neuron i
ξ^μ	Stored pattern μ ($\mu = 1, \ldots, P$)
J_{ij}	Hopfield weight from j to i
h_i	Local field (total recurrent input)
m^μ	Overlap with pattern μ
$\alpha = P/N$	Memory load

Hebbian storage: $J_{ij} = \frac{1}{N} \sum_{\mu=1}^{P} \xi_i^\mu \xi_j^\mu, \quad J_{ii} = 0$

Update rule: $s_i \leftarrow \text{sign}\Big(\sum_j J_{ij} s_j \Big)$

Overlap: $m^\mu = \frac{1}{N} \sum_i \xi_i^\mu s_i$

Energy: $E(\boldsymbol{s}) = -\frac{1}{2} \sum_{i,j} J_{ij} s_i s_j$

Capacity (random dense patterns): $P_{\max}/N \approx 0.138$